MIL-F-43310A
30 December 1970
SUPERSEDING
MIL-F-43310
12 February 1965

MILITARY SPECIFICATION

FELT SHEET: NYLON, NON-WOVEN (NEEDLE PUNCHED)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers nylon felt intended for use as padding for laundry presses.

1.2 Classification.- The felt shall be of the following types as specified (see 6.2).

Type I - 13.0 ounces per square yard.
Type II - 18.0 ounces per square yard.
Type III - 29.0 ounces per square yard.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	-	Twine, Cotton, Wrapping.
T-T-911	-	Twine, Jute
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
CCC-C-429	-	Cloth, Osnaburg, Cotton
CCC-C-467	-	Cloth, Burlap, Jute (or Kenaf).
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing.

FSC 8305

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MILITARY

MIL-L-10547 - Liners, Case and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-T-40625 - Tubing, Bias Sewn, Burlap or Osnaburg Cloth.

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

* 2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Attn: Tariff Order Section, 1616 P. Street, N.W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Material.-

3.1.1 Felt.- The felt shall be a non-woven (needle punched) construction and shall be made of nylon of suitable length and denier to comply with the requirements of this specification. The nylon shall have a melting point of (482° ± 10°F). The use of regenerated or reprocessed nylon is prohibited (see 4.4).

3.2 Color.- The color of the finished felt shall be natural.

3.3 Physical requirements.- The finished felt shall conform to the physical requirements specified in table I when tested as specified in 4.4.

TABLE I. - Physical requirements

Type	Weight per square yard ounces (min.)	Thickness inches (min.)	Breaking strength pounds (min.) Length	Width	Compression deformation percent	Resilience percent
					Maximum	Minimum
I	13.0	0.130	10	10	32.0	70.0
II	18.0	0.180	10	10	32.0	70.0
III	29.0	0.310	40	40	32.0	70.0

3.3.1 Width.- The finished width shall be 54 inches minimum.

3.4 Finish.- The staple or felt shall be heat treated. The finished felt shall show not more than 3.0 percent dimensional change in either length or width when tested as specified in 4.4.

3.4.1 Accelerated ageing.- The finished felt shall not lose more than 50 percent of the original breaking strength when tested as specified in 4.4.

3.5 Length and put-up - Unless otherwise specified (see 6.2), the felt shall be in continuous lengths, each not less than 25 continuous yards. Each length shall be put up on rolls.

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3.6 Identification tickets.- Unless otherwise specified, each roll of felt shall have a ticket attached to the edge at the inner end of the felt with not finer than 5-ply cotton string doubled to not less than 8 inches long. The ticket shall be made of not less than 20-point paper stock, and the color shall be white or light in intensity to permit easy reading of printed, stamped or typed markings. Entries on tickets shall be printed, stamped or typed. Handwritten entries are prohibited. The ticket shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water-soluble ink with the following information:

Nomenclature----------
Type---------Thickness----------
Federal Stock Number----------
Supplier----------
Contract number and date----------
Specification number and date----------
Name of contracting office----------
Supplier's piece number----------
Net yards----------

3.7 Fiber identification - Each roll of felt shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.8 Workmanship.- The finished felt shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter

* 4.2 1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Examination of the end item.- Examination of the felt shall be in accordance with 4.3.1 through 4.3.3.2.

4.3.1 Yard-by-yard examination.- The required yardage of each roll shall be examined on the face side and defects classified in accordance with the list below. A continuous defect shall be counted as one defect for each lengthwise yard or fraction thereof in which it occurs. The sample unit shall be 1 linear yard of felt. The sample size shall be based upon inspection level II of MIL-STD-105. The sample size and the lot size shall be expressed in units of 1 yard each. The number of rolls from which the sample is to be selected shall be in accordance with table II of this specification. An approximately equal number of yards shall be examined in each roll of the sample. The acceptable quality level (AQL) shall be 6.5 (one class) defects per hundred units (yards).

Defects

Any hole, cut, tear.
Any spot, stain or streak - clearly visible 1/
Crease or wrinkle, hard embedded.

1/ At the normal inspection distance (approximately 3 feet).

TABLE II.-

Lot size in yards	Sample size in rolls	Maximum no of defects acceptable in samples
Up to 1200 1/	3	0
1201 up to and including 3200	5	0
3201 up to and including 10,000	7	0
10,001 up to and including 35,000	10	0
35,001 up to and including 150,000	15	0
150,001 and over	25	1

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

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4.3.2 Overall examination.- Each defect listed shall be counted not more than once in each roll examined. The sample unit shall be one roll. The sample size (number of rolls selected as sample) and the number of defects acceptable shall be as shown in table II.

Overall defect
Width less than specified.

4.3.3 Examination for length -

4.3.3.1 Individual roll.- The roll shall be examined for length. Any length found to be less than the specified minimum length, or more than 2 yards less than the length marked on the ticket or not in a continuous length shall be scored a defect. The sample unit shall be one roll. The number of rolls selected as sample (sample size) and the acceptance number for this examination shall be in accordance with table II.

4.3.3.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the tickets. The rolls examined shall be those selected for examination of individual rolls.

4.3.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking complies with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 of MIL-STD-105 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted: incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.

* 4.4 Tests.- The methods of testing specified in FED-STD-191, wherever applicable and as specified in table III shall be followed. The physical and chemical values specified in Section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method, except for breaking strength initially and after accelerated ageing.

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The breaking strength requirements are applicable to the average of the determinations made on all sample units (lot average). The sample unit shall be 1-1/2 continuous yards, full width of the finished felt. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requriement specified or if the breaking strength results based on the average of all determinations (lot average) fail to meet the specified requirement. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size yards	Sample size (units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. - Test methods

Characteristic	Requirement Paragraph	Test method
Material		
Construction 1/	3.1.1	Visual
Melting point	3.1.1	1534 2/
Regenerated or reprocessed nylon	3.1.1	2/
Weight	3.3	5041
Thickness	3.3	5030 and 4.4.1
Compression deformation	3.3	4.4.1
Resilience	3.3	4.4.1
Breaking strength:		
Initially	3.3	4.4.2
After heat ageing	3.4.1	4.4.3
Finish:		
Heat treatment	3.4	2/
Dimensional stability	3.4	4.4.4

1/ One determination per sample unit shall be reported as "pass" or "fail".

2/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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4 4.1 The thickness, compression deformation, and resilience shall be determined in accordance with Method 5030 and as follows, except that the diameter of the supporting anvil shall be a minimum of 2 inches:

4 4.1.1 Thickness shall be determined as specified in Method 5030 and recorded as "A". Allow presser foot to remain in contact with the same material area and immediately subject the specimen to an additional load of 2 pounds per square inch (p.s.i.) for 60 seconds. At the end of the 60 second period record the thickness as "B", the compressed thickness of the specimen. Retain the pressure foot in contact with the material and immediately remove the 2 p.s.i. load. After 60 seconds record the specimen thickness as "C"

4.4 1.2 The above procedure shall be randomly repeated five times over the surface of the sample unit. The individual results shall be reported to the nearest 0.001 inch, averaged to obtain an average sample unit value for A, B and C and reported to the nearest 0.001 inch. These average values shall be utilized in calculating compression deformation and resilience in accordance with the formula below, and reported to the nearest 0.1 percent. The thickness shall be recorded as the average of the A values.

$$\text{Compression deformation, percent} = \frac{A-B}{A} \times 100$$

$$\text{Resilience, percent} = \frac{C}{A} \times 100$$

4.4.2 Determination of initial breaking strength.- The breaking strength shall be determined in accordance with Method 5100 except that 2 by 6 inch specimens shall be cut with a die from each sample unit, 5 parallel to the lengthwise direction of the cloth and 5 parallel to the width of the cloth. All single determinations for each direction from all sample units shall be averaged and the averages reported as to the breaking strength of the width and length as applicable.

4.4.3 Determination of breaking strength after heat ageing.- Ten specimens prepared as specified in 4.4.2 shall be exposed in a circulating oven for 5 hours at a temperature of 350° ± 5°F. The specimens shall be cooled and then allowed to reach equilibrium under Standard Conditions prior to determining breaking strength as specified in 4.4.2. All single determinations for each direction from all sample units shall be averaged. The results shall be reported as the breaking strength after accelerated ageing for the applicable cloth direction.

4.4.3.1 The percent loss in breaking strength shall be determined for each felt direction in accordance with the following formula and results reported to the nearest 1 percent.

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$$\text{Loss in breaking strength, percent} = \frac{O-A}{O} \times 100$$

Where O = Initial breaking strength (lot average)
A = Breaking strength after accelerated ageing (lot average)

4.4.4 Dimensional stability.- Each test specimen shall be a square of felt at least 12 by 12 inches. Each specimen shall be laid flat without tension, and marked off with a square 10 by 10 inches, having sides parallel to the length and width directions of the felt. The test specimens shall be placed in an oven maintained at 280°F. (± 5°F.) for a period of two hours. At the end of this period, the specimens shall be removed from the oven, and conditioned under Standard Atmospheric Conditions. The specimens shall then be measured for changes in dimensions. Three specimens shall be tested from each sample unit. Changes in dimensions shall be reported to the nearest 0.1 percent for both the length and width directions.

* 5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Felt shall be wound open width on a convolute chipboard tube. The ends of the tube shall be flush with, or extend not more than 1 inch beyond each side of the width of the rolled felt. Each roll shall be completely inclosed in two thicknesses of 60-pound basis weight kraft paper conforming to grade B of UU-P-268. The paper wrapping shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade C of PPP-T-45.

5.1.2 Level C.- Felt shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing - Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.- Each roll of felt, packaged as specified in 5.1 shall be completely overwrapped in a waterproofed flexible sheet conforming to type III, grade B or C of MIL-L-10547. The seams and folds of the overwrap sheet shall be secured by one of the methods specified in the appendix of the overwrap sheet specification. Each overwrapped roll of felt shall be wrapped and sewn in cotton osnaburg cloth conforming to class 2 of CCC-C-429 or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The wrapping shall be securely sewn with 16-ply cotton twine conforming to T-T-871

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or jute twine conforming to type I, size 2 of T-T-911, with approximately one stitch to the inch and every third stitch knotted. Five inches of surplus covering shall be gathered together on each of the four corners and securely sewn into ears for handles. In lieu of the sewn wrapping, the cloth covering may be in bias-sewn tubing form or bags (tubing presewn one end) conforming to MIL-T-40625, with closure effected by securing each open end with two wire ties. The first wire tie shall be applied as close to the contents as possible. The second wire tie shall be applied approximately 1 inch from the first wire tie, with the twisted ends positioned opposite to the ends of the first wire tie. The wire ties shall be not less than 6 inches long, 0.072-inch thick galvanized soft iron or steel wire with a 1/2-inch diameter formed eye at each end.

5.2.2 Level B.- Each roll of felt, packaged as specified in 5.1, shall be wrapped in cloth covering as specified in 5.2.1. The waterproofed barrier material overwrap shall not be required.

5.2.3 Level C.- The felt, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipments shall be in accordance with Uniform Freight Classification rules or National Motor Freight rules as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The felt covered by this specification is intended for use as padding for laundry presses.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Length required, if other than specified (see 3.5).
(d) Selection of applicable levels of put up, packaging and packing (see 5.1 and 5.2).

6.3 Marginal notations.- The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections) from the previous issue were made. This was done as a convenience only and

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Army - GL
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Air Force - 82

Preparing activity:

Army - GL

Project No. 8305-0735

Review activities:

Army - MD

User activities:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Felt Sheet: Nylon, Non-Woven (Needle Punched) MIL-F-43310A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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